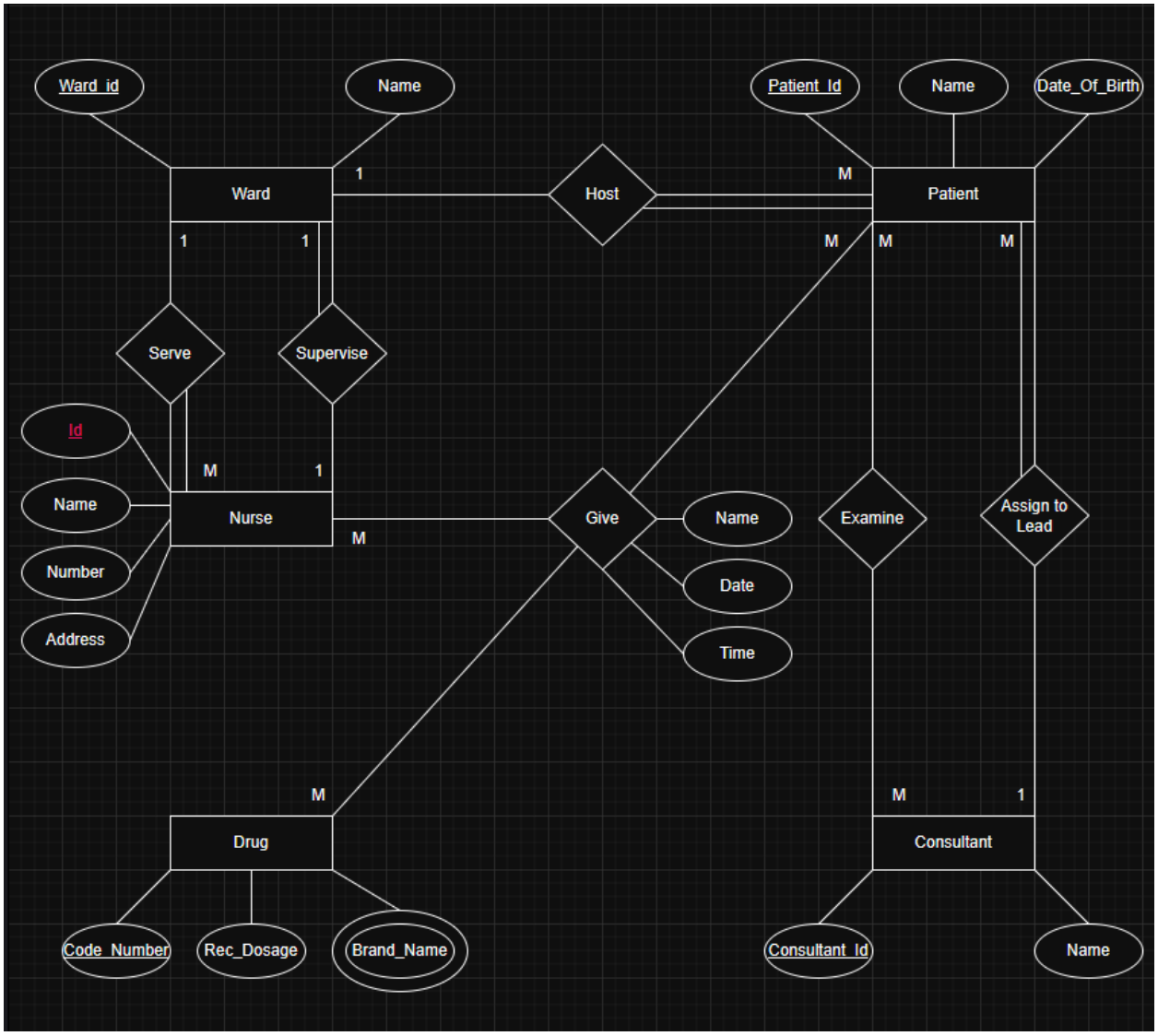
ERD Examples:

- A General Hospital consists of a number of specialized wards. Each ward is described by ward_id, Name
- The system records the following details about patients: Patient_id, name, Date_Of_Birth
- Each ward may host one or more patients and each patient is hosted by only one ward.
- Each patient is assigned to one leading consultant but may be examined by other consultants, if required.
- Each consultant may be assigned zero or more patients and may examine zero or more patients.
- Consultants are described by Consultant_id, Name
- The system has to record all required data each time the Nurse gives a patient a certain drug with specified dosage at certain date and time.
- Each ward is under supervision of one nurse and a nurse may supervise only one ward.
- Each Nurse must serve in one ward
- Data about the nurse is recorded as her name and her number and her address.
- A drug has code number, recommended dosage and more than one brand name



Major airlines companies that provide passenger services keep database with lots of information on all airlines.

1. Each airline has an identification number, name and address, name of the contact person and telephone numbers.

2. Each employee works in Airline Company has an employee identification number, name, address, birthday recorded as (day, month, year), gender, position with the company, and qualifications.

3. Each airline owns different aircraft models. For each aircraft an aircraft identification number, capacity, and model is recorded.

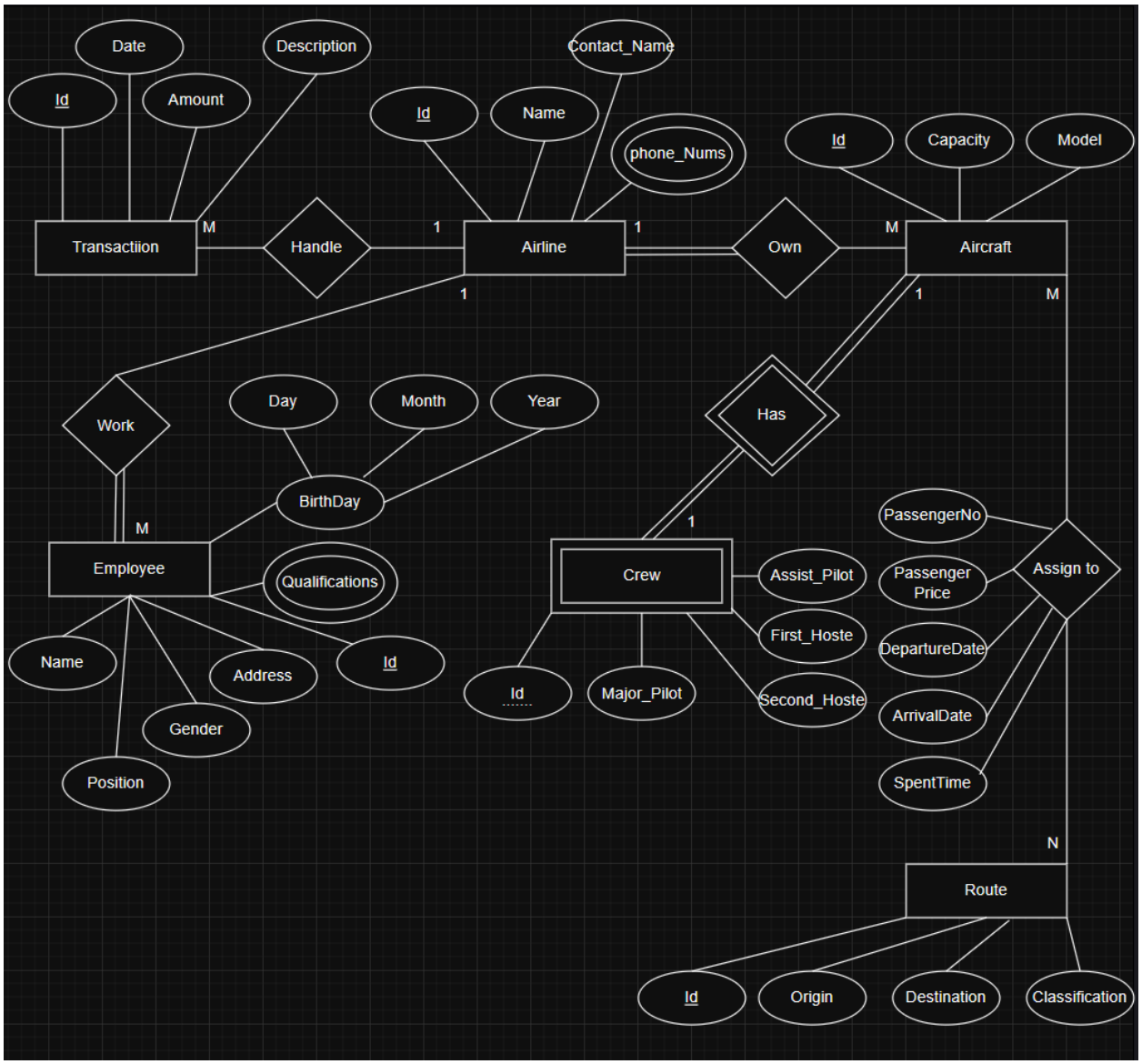
4. The aircrafts are assigned to different routes. An aircraft can work on more than one route and a route has many aircrafts going on it. Some information as number of passengers, price per passenger, departure date time, arrival date time and the time that aircraft spent in travelling the route are recorded.

Each route has a route identification number, origin, destination, distance,classification (e.g. domestic or international route).

5. Each aircraft has its own crew (major pilot, assistant pilot and two hostesses), the aircraft crew not stored as employee. Each crew is assigned to one aircraft.

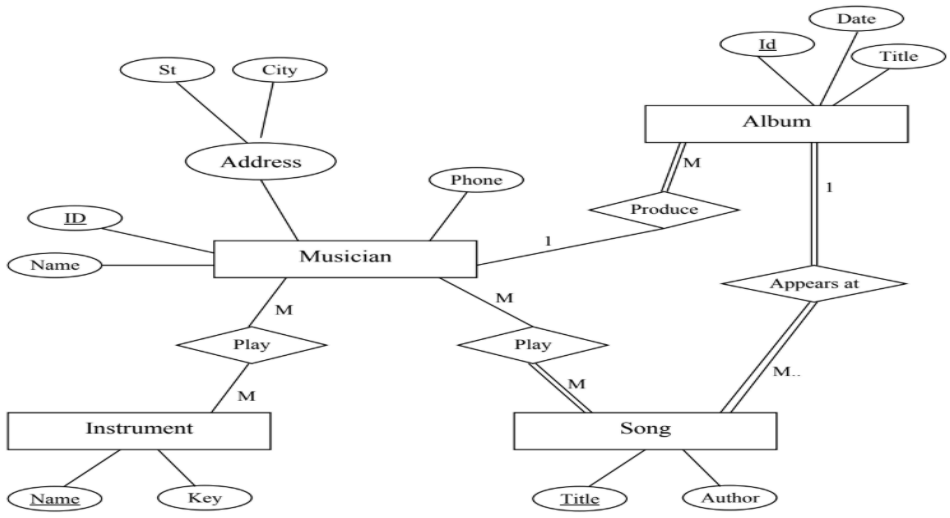
6. Each airline keeps information about their buy/sell transactions (for example selling an airplane ticket is a sell transaction, paying for maintenance is a buy transaction). Each transaction has a transaction identification number, date, description, and amount of money paid/received.

Draw an E-R diagram for the database presented above.



Mapping the ERD

First ERD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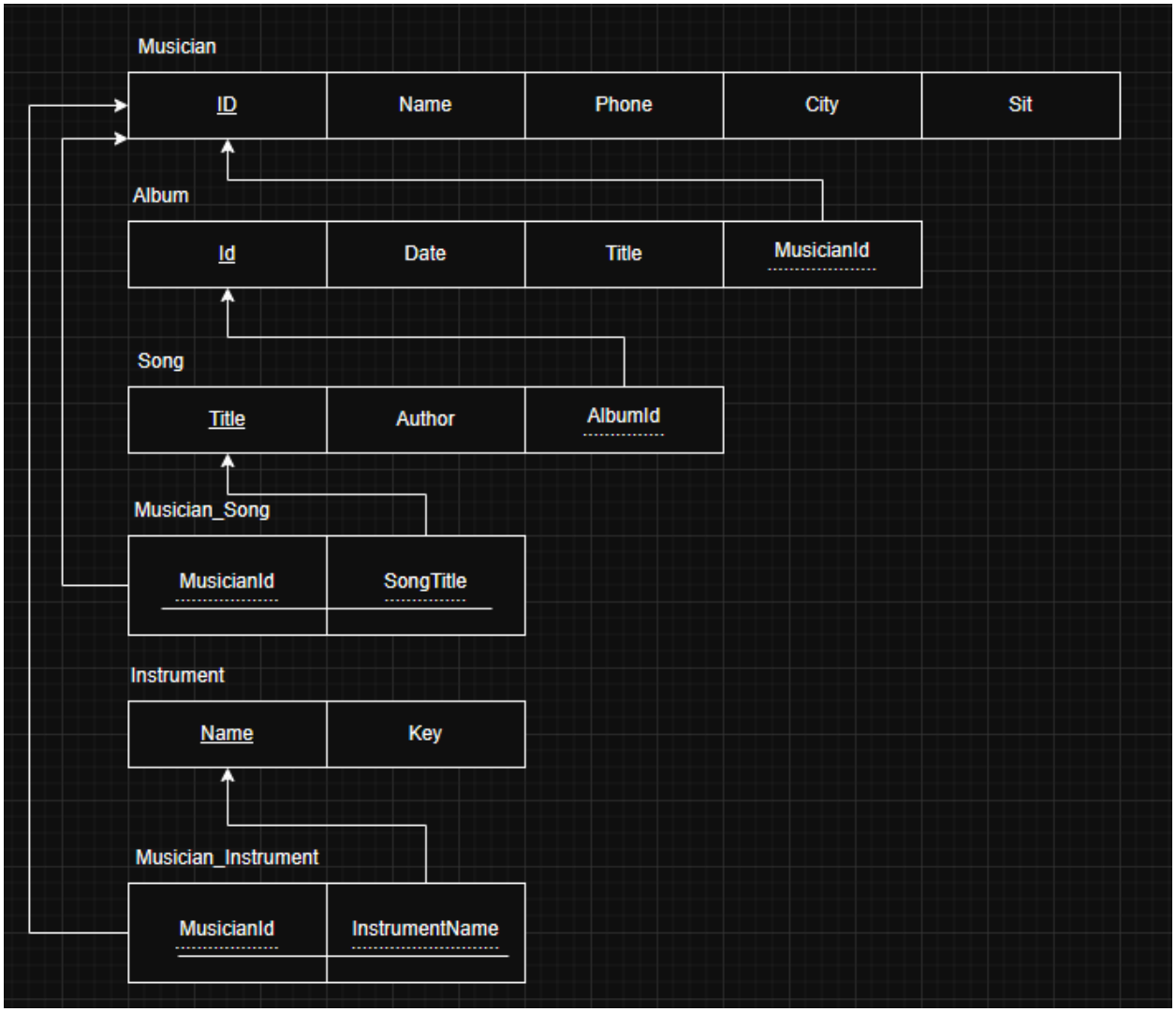
Id
Title
St
City
Album
Address
M
Phone
1
Produce
ID
Musician
1
Name
Appears at
M
M
Play
Play
M..
M
M
Instrument
Song
Name
Key
Title
Author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Phone
City
Sit
Album
Id
Date
Title
MusicianId
Song
Title
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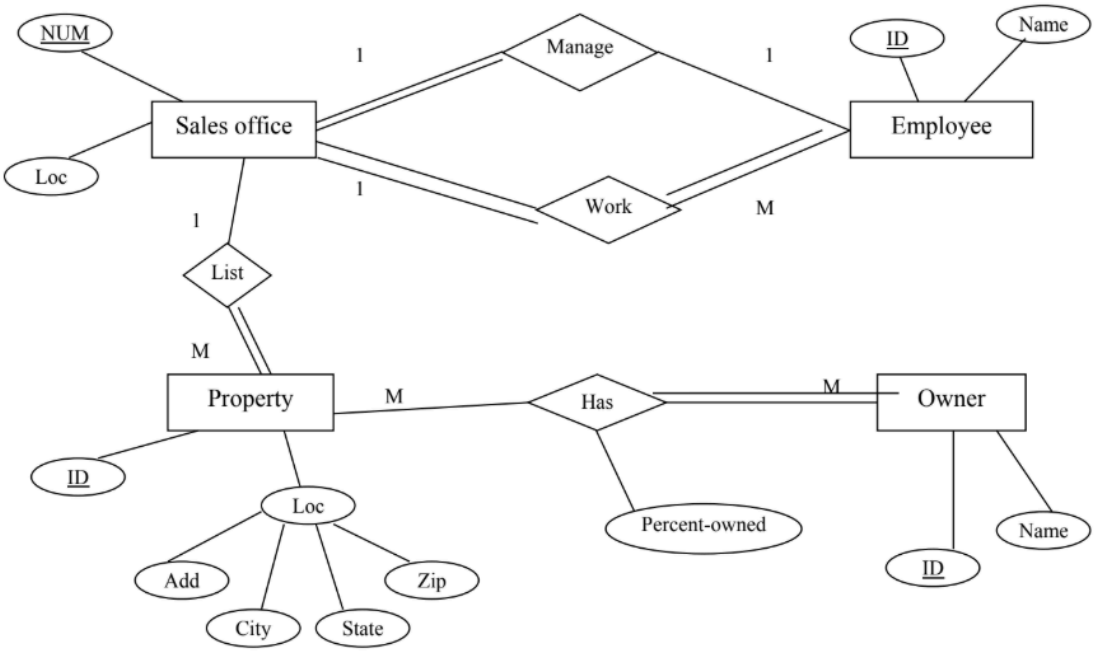
SongTitle
Instrument
Name
Key
Musician_Instrument
MusicianId
InstrumentName

Second ERD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Manage
1
ID
Name
Employee
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Work
M
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List
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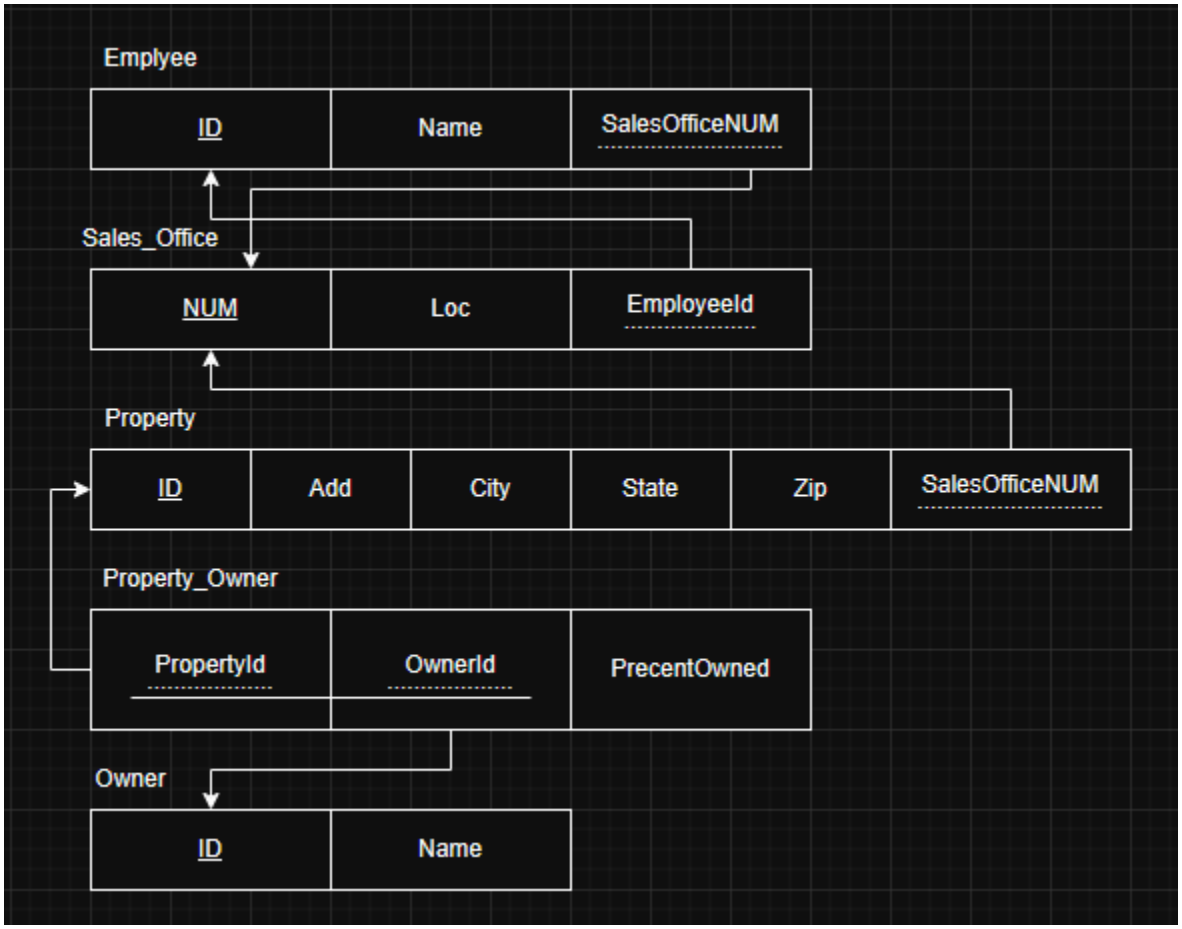
Property
M
Has
M
Owner
ID
Loc
Add
City
State
Zip
Percent-owned
Name
ID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SalesOfficeNUM
Sales_Office
NUM
Loc
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Property
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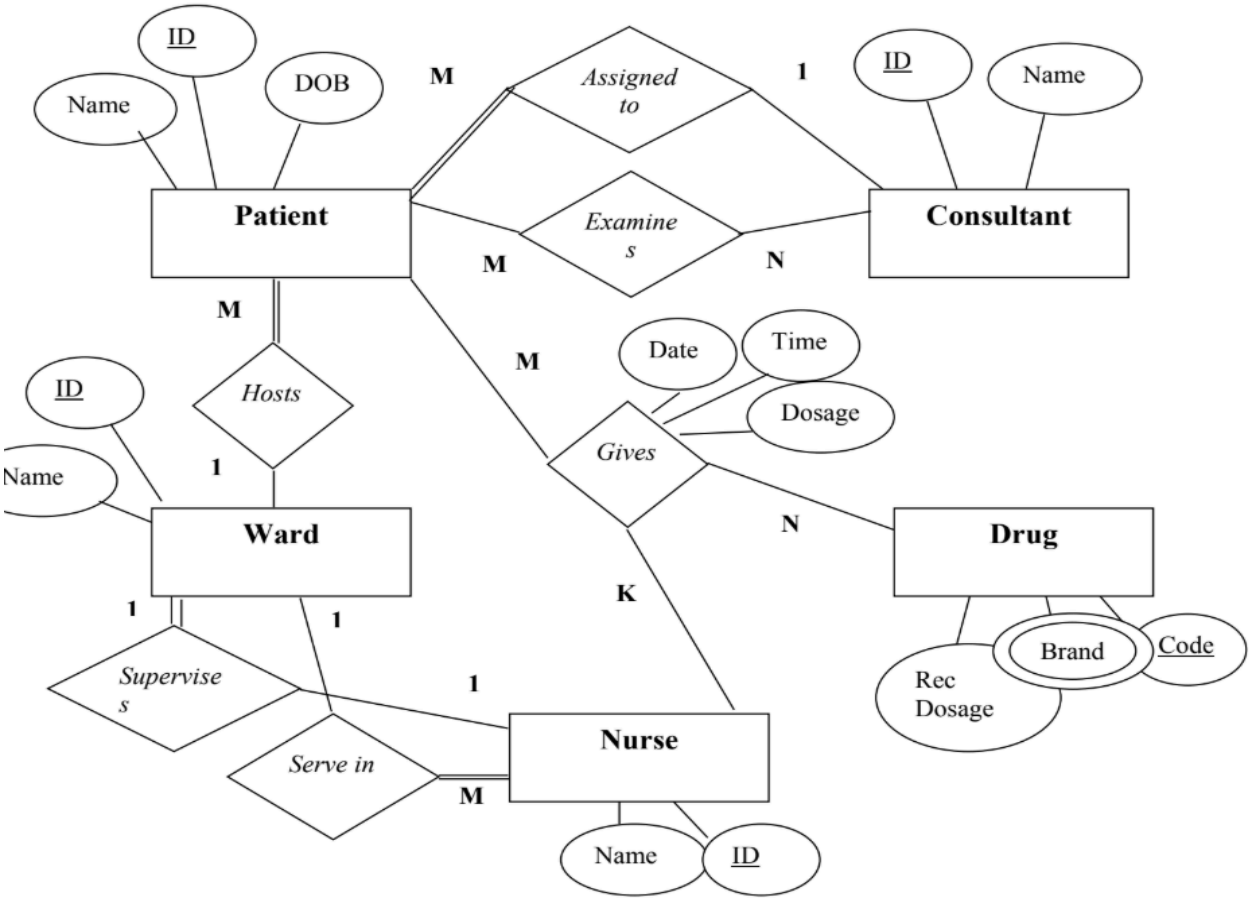
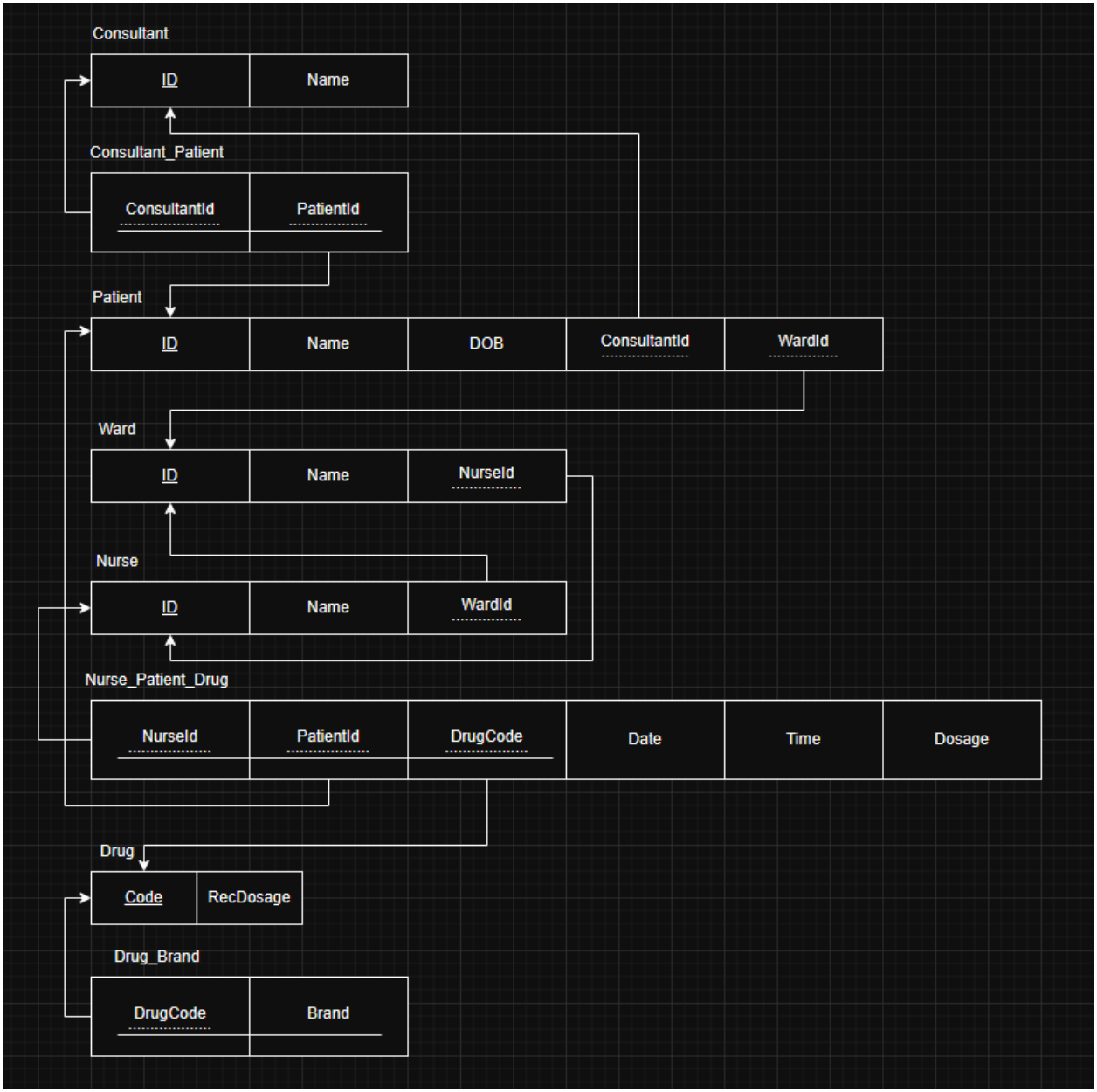
Add
City
State
Zip
SalesOfficeNUM
Property_Owner
PropertyId
OwnerId
PrecentOwned
Owner
ID
Name

Third ERD


ID
Name
DOB
M
Assigned to
1
ID
Name
Patient
Examines
Consultant
M
N
M
M
Date
Time
Dosage
ID
Hosts
Gives
Name
1
Ward
N
Drug
K
1
1
Brand
Code
Supervises
1
Rec Dosage
Serve in
Nurse
M
Name
ID


Consultant
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Name
Consultant_Patient
ConsultantId
PatientId
Patient
ID
Name
DOB
ConsultantId
WardId
Ward
ID
Name
NurseId
Nurse
ID
Name
WardId
Nurse_Patient_Drug
NurseId
PatientId
DrugCode
Date
Time
Dosage
Drug
Code
RecDosage
Drug_Brand
DrugCode
Brand